

Recall **Theorem 14**: Let A be an ϵ_o -optimizer for the empty history with either (1) the ability to self-modify and modification-independent utility function and belief, or (2) without the ability to self-modify and with a possibly modification-dependent utility function and belief. Let γ be the agent's discount rate, ϵ_u the error in utility function with respect to the correct utility function u^* and ϵ_ρ the absolute error in belief with respect to the correct belief ρ^* . Then at timestep t :

- (1) If we let ϵ' be smallest such that A at time t is an ϵ' -optimizer, then $E_{\mathcal{H}_{<t}}[\epsilon'] \leq f_{\text{opt}}(\epsilon_o, \gamma) + f_{\text{util}}(\epsilon_u, \gamma) + f_{\text{bel}}(\epsilon_\rho, \gamma) + f_{\text{disc}}(\gamma, \gamma^*)$ where the expectation is over histories where percept are distributed according to ρ^* and actions are given by the agent's policy. Moreover, if $\epsilon_o = 0$, then $\epsilon' \leq f_{\text{opt}}(\epsilon_o, \gamma) + f_{\text{util}}(\epsilon_u, \gamma) + f_{\text{bel}}(\epsilon_\rho, \gamma) + f_{\text{disc}}(\gamma, \gamma^*)$ almost surely.
- (2) A will be an ϵ' -optimizer, with respect to the correct discount rate γ^* , where $\epsilon' \leq \epsilon_o + f_{\text{util}}(\epsilon_u, \gamma) + f_{\text{bel}}(\epsilon_\rho, \gamma) + f_{\text{disc}}(\gamma, \gamma^*)$

Moreover, when $\gamma \geq 1/2$, there exists an agent which achieves equality up to a factor of at most 8 and up to a factor of 16 if ϵ_ρ is a relative error.

Proof.

Upper bound: We prove that in both cases $\epsilon' \leq \epsilon_{o,t} + f_{\text{util}}(\epsilon_u, \gamma) + f_{\text{bel}}(\epsilon_\rho, \gamma) + f_{\text{disc}}(\gamma, \gamma^*)$ with $\epsilon_{o,t}$ being the optimization guarantee of the agent at timestep t . From Theorem 7, we have that for modification-independent belief and policy, if we start with an ϵ -optimizer, then at step t we have $E_{\mathcal{H}_{<t}}[\epsilon_{o,t}] \leq \min(\frac{\epsilon}{\gamma^{t-1}}, \frac{1}{1-\gamma})$. In the case of non-self-modifying agents, the optimization guarantee stays the same.

Let $A_0, \dots, A_5$ be agents such that A_0 is a perfect optimizer with perfect knowledge, A_1 is the same as A_0 except it has an inaccurate discount factor γ , A_2 has additionally inaccurate utility function with error ϵ_u , A_4 has additionally absolute error of ϵ_ρ in belief and A_5 has additionally ϵ_o -optimizing policy. Then

$$|V_{A_0}(\mathcal{H}_{<t}) - V_{A_5}(\mathcal{H}_{<t})| \leq \sum_{i=1}^4 |V_{A_i}(\mathcal{H}_{<t}) - V_{A_{i+1}}(\mathcal{H}_{<t})|$$

In this sum, the summands can be bounded by $f_{\text{disc}}(\gamma_u, \gamma_o)$, $f_{\text{util}}(\epsilon_u, \gamma_u)$, $f_{\text{bel}}(\epsilon_\rho, \gamma_u)$, and $\epsilon_{o,t}$ respectively thanks to theorems 9, 11, and 13.

The bound in the case (2) follows directly. In the case of (1), this gives us

$$E_{\mathcal{H}_{<t}}[\epsilon'] \leq E_{\mathcal{H}_{<t}}[\epsilon_{o,t} + f_{\text{util}}(\epsilon_u, \gamma) + f_{\text{bel}}(\epsilon_\rho, \gamma) + f_{\text{disc}}(\gamma, \gamma^*)] = f_{\text{opt}}(\epsilon_o, \gamma) + f_{\text{util}}(\epsilon_u, \gamma) + f_{\text{bel}}(\epsilon_\rho, \gamma) + f_{\text{disc}}(\gamma, \gamma^*)$$

Lower bound: We show the bound in the case of absolute error in belief, the proof for the relative error is analogous. One of $f_{\text{opt}}, f_{\text{util}}(\epsilon_u, \gamma_u), f_{\text{bel}}(\epsilon_\rho, \gamma_u), f_{\text{disc}}(\gamma_u, \gamma_o)$ has to be greater or equal to $\epsilon'/4$. Let this be the case for $f_\bullet$ for $\bullet \in \{\text{opt}, \text{util}, \text{bel}, \text{disc}\}$. We know from theorems 9, 11 and 13 that there exists an agent such that $\epsilon'_\bullet$ is tight up to a factor of 2 (for bounded-optimization agent, we are using that $\gamma \geq 1/2$). For this agent, ϵ' is then tight up to a factor of 8. $\square$

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